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Silicon Export from Small Watersheds

Introduction

Chemical weathering is the process by which rock is dissolved into water. Chemical weathering of silicate rocks is a key control on the earth climate system over geologic time (Torres et al, 2007). However, modern weathering rates are poorly understood. Research shows that acidity (Meybeck, 1987) , bedrock material (Meybeck, 1987), water availability (Gaillardet et al, 1999), and plant growth (Lyons et al, 2005) are all key drivers to changing weathering rates. Since the passage of the Clean Air Act Amendments of 1990 (CAAA), acid deposition rates across the United states have fallen to the lowest levels on record (Stoddard et al, 1999). Whether or not this has had a significant effect on modern weathering rates, and how this will continue to change in the future, are not well understood.

One way to proxy weathering rates is to look at the export of weathering products from rivers (Gaillardet et al, 1999). Solute concentrations in rivers represent an integration of processes that occur in their surrounding watershed (Maher and Chamberlain, 2014). Our investigation seeks to use several modeling techniques to test how the passage of the CAAA have changed drivers of chemical weathering in the United States.

Methods

Data Selection and Aggregation

Data was sourced from Macrosheds. We selected three sites from the Macrosheds dataset based on their geographic diversity, undeveloped status, silicate bedrock, and data availability. We chose to use data from a site at H.J. Andrews Experimental Forest in California, Hubbard Brook Experimental Forest in Vermont, and Niwot Ridge Critical Zone Observatory in Colorado. All sites had data on silicon concentration, streamflow, precipitation, stream water acidity, and in watershed gross primary production. These data spanned from before the CAAA to present day.

Load was calculated by linearly interpolating the silicon data to the daily streamflow data, and then calculating daily flux by multiplying the two together and then summing total daily silicon export by each month. Likewise, the other parameters were interpolated to the daily timestep and then aggregated to the monthly timestep.

Data was then split into two groups, pre-CAAA and post-CAAA. Monthly data spanning from the 1988 and 1989 water years was designated as the pre-CAAA dataset. Data spanning the 2011 and 2012 water years was designated as the post-CAAA dataset. Each dataset contains 72 site-month observations.

Variable Selection

As noted above, past research indicates several variables are known to be useful in determining silicon load in waterways. However, we underwent variable selection to confirm these variables were useful in our models. Initially in the dataset there were 26 variables. Several of these variables had missing data, the same data, or duplicate data. These variables were removed from the dataset. The remaining variables were used in a correlation heatmap (figure 1).

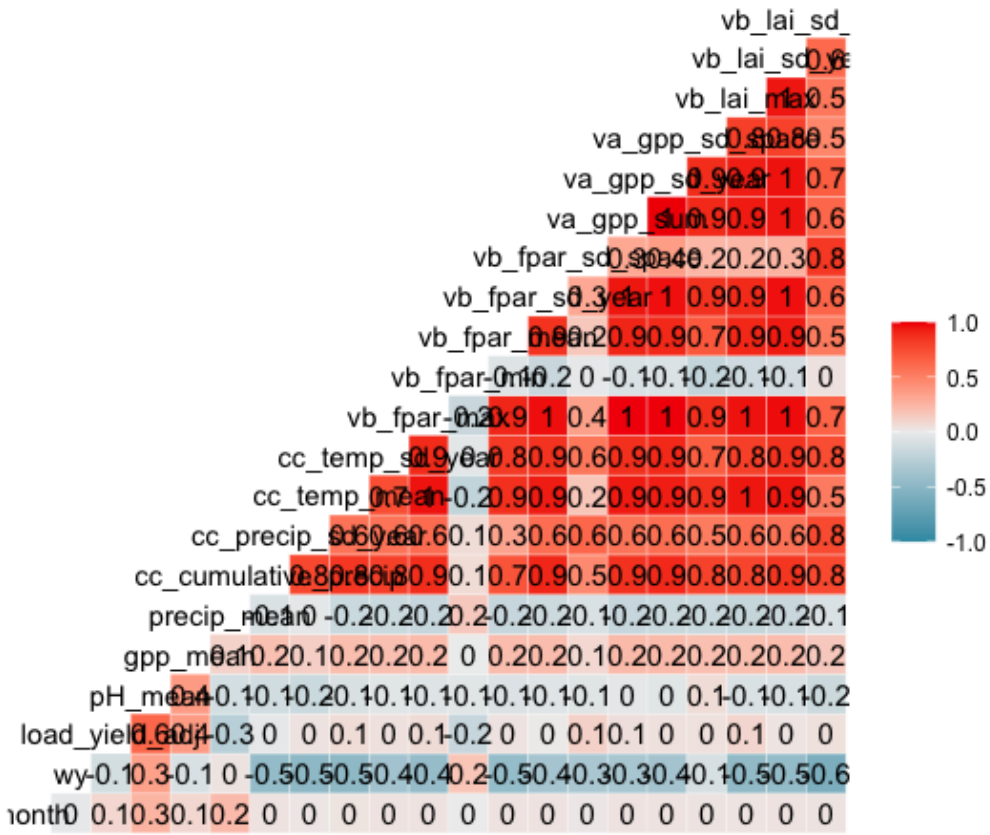

Figure 1: A correlation heatmap after first variable selection.

There are several variables that are highly correlated with each other but not our target variable: load yield adjusted. These variables non-correlated with our target predictor were removed and the remaining variables were put in another correlation heatmap (figure 2).

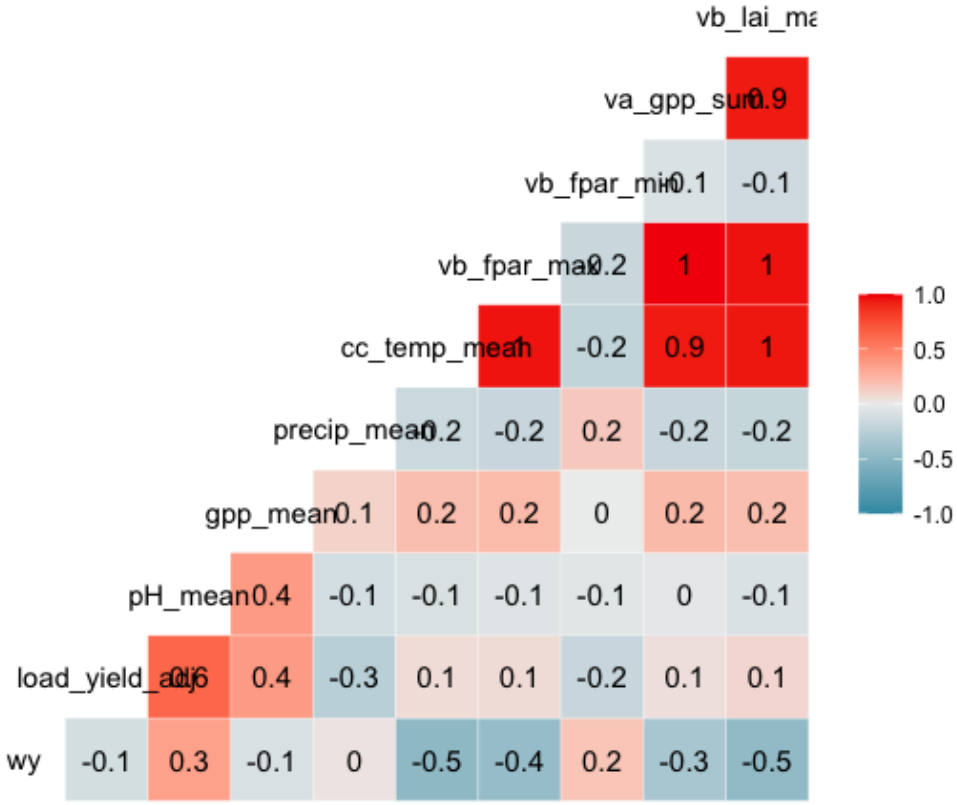

Figure 2: Correlation heatmap after second variable selection

The variables we chose to include in our models are yield, pH, GPP, precipitation, site, and month. A final correlation plot was produced from our chosen variables for both pre-CAAA and post-CAAA datasets (figures 3 and 4).

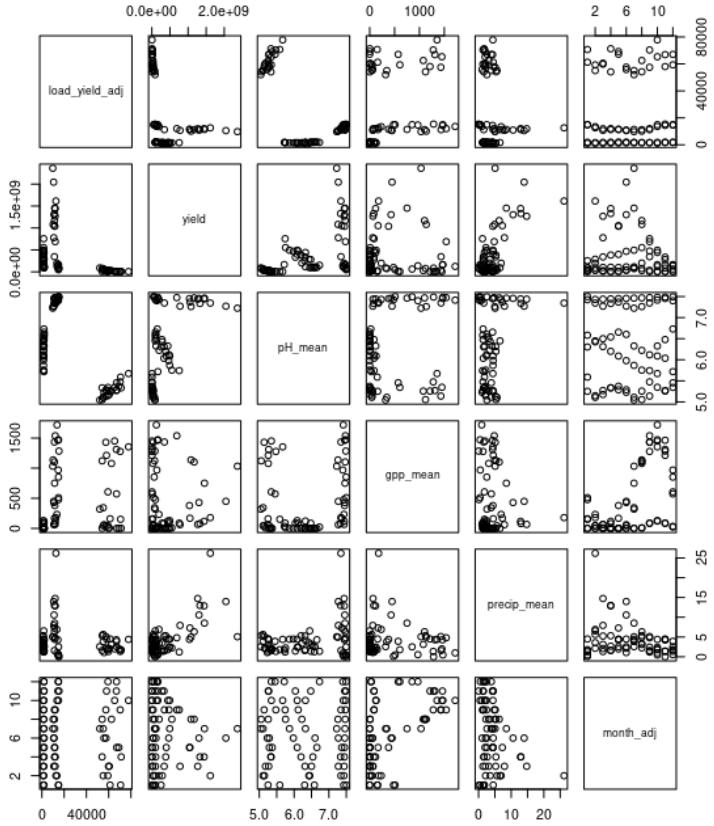

Figure 3: Correlation plot of variables used in our investigation of pre-CAAA models.

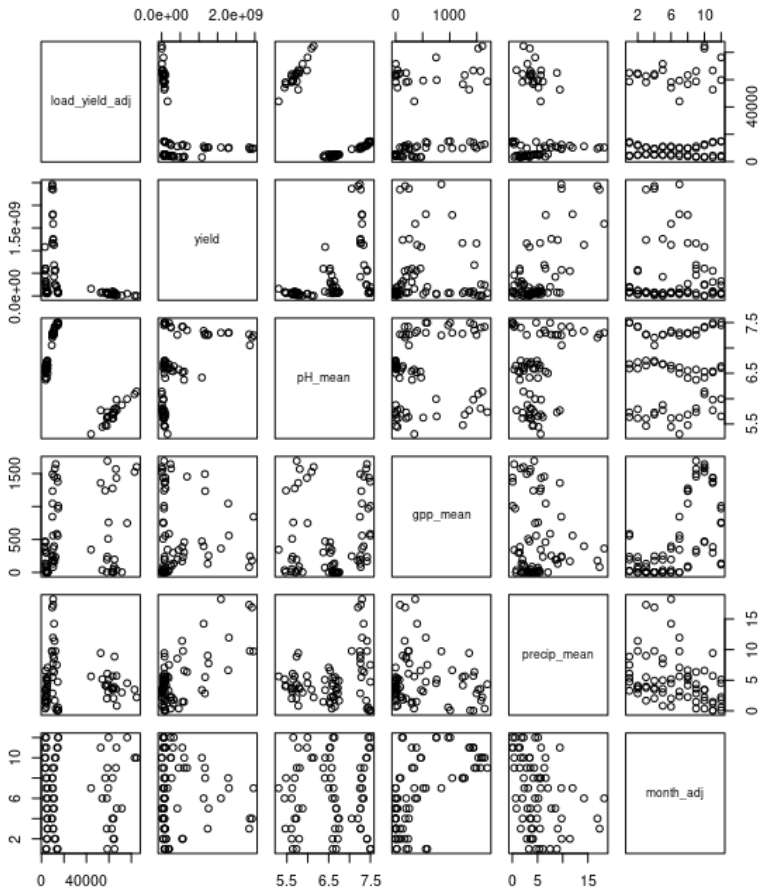

Figure 4: Correlation plot of variables used in our investigation of post-CAAA models.

MLR

The multiple linear regression was used for the starting method due to its relative parsimony. Using a simple model as a starting point made it maximally interpretable. Using the variables selected we included all predictors in a MLR and ran this on both the pre-CAAA and post-CAAA data sets. We also subsetted the data into each site to get a sense of site specific effects.

GLM

In order to compare how drivers of Silicon export have changed from before and after the CAAA, we fit generalized linear models to our pre-CAAA and post-CAAA datasets and compared the coefficient estimates. In order to select a value for lambda in our ridge regression, we tuned our lambda term with 10-fold cross validation. The values used for lambda ranged from 0.01 to $1x10^{10}$. The value of lambda that gave the lowest root mean square error (RSME) was selected for each final model.

Random Forest

Two methods of random forest were used to assess prediction for Pre-CAAA and Post-CAAA silicon load. Bagging was the first method used with parameter 'mtry' (or number of variables used in the model) set to 7 (the maximum number of variables to be included from our data). Additionally the default of 500 trees produced was used for bagging. The second method was gradient boosting with 10 fold cross validation. The training set was set to 80% and the testing 20% for both methods.

Results

MLR

After running the Multiple Linear regression we found that overall the only significant predictor in the combined pre CAAA data set was pH, we felt that this was skewed with a bias on the sites in the data set and as we mentioned above we subsetted by site, This is shown in figures 5 and 6.

Figure 5: The predicted outcome of the yield load adjusted on the x axis and the real yield load adjusted with the site of the data the color of the data point. The data used for this plot is the pre CAAA data collected.

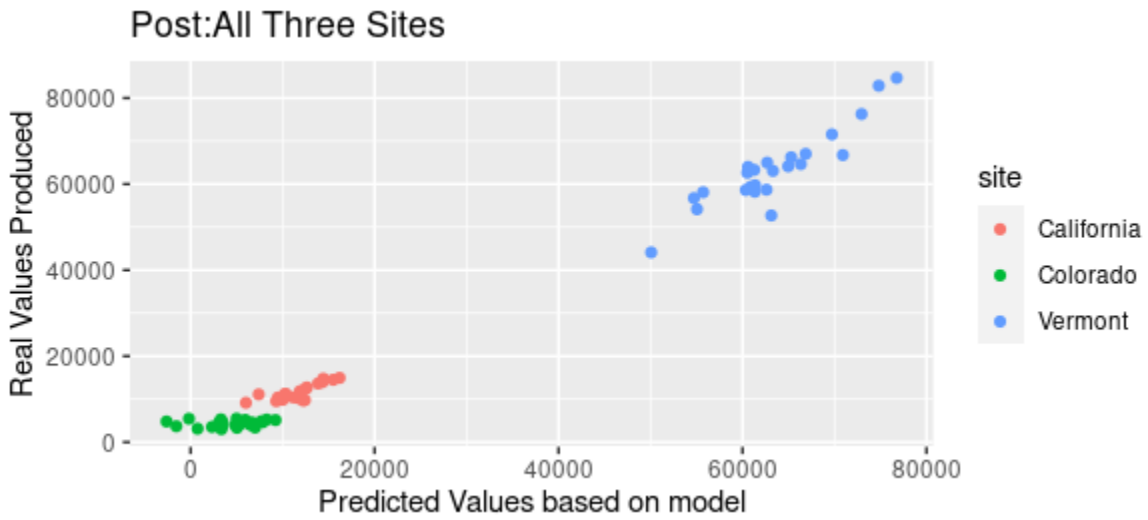

Figure 6: The predicted outcome of the yield load adjusted on the x axis and the real yield load adjusted on the y axis, with the site of the data the color of the data point. The data used for this plot is the post-CAAA data.

Looking at the pre-CAAA data set in Colorado, there are no significant predictors in estimating the adjusted load. We repeated the process for California and Vermont, California can back with total yield as the most important predictor which is contrary to Vermont which came back with pH as the sole significant predictor. Vermont is consistent with the entire pre-CAAA data set but the other two sites did not have the same predictors.

In the post-CAAA dataset of all sites the two significant predictors are pH and yield, the coefficients we found changed from 8070 in the pre-CAAA dataset to 33200 in the post-CAAA dataset. These changes are shown in table 1.

Table 1: A table of coefficient estimates from the multiple linear regression model. The table Includes the terms of the variables and the coefficients of those variables from the pre-CAAA and post-CAAA data sets from all sites.

Term	Estimate Pre-CAAA	Estimate Post-CAAA
Intercept	47900	-233000
Yield	-0.000000612	0.00000308
pH	8070	33200
GPP	0.863	-1.58
Precipitation	-16.6	-282
Month of Water Year	93.4	111
Site Effect Colorado	-1350	17700
Site Effect Vermont	66300	108000

When isolating the MLRs by site, in the post-CAAA dataset Colorado does have a single significant predictor, pH, as visible in figure 7.

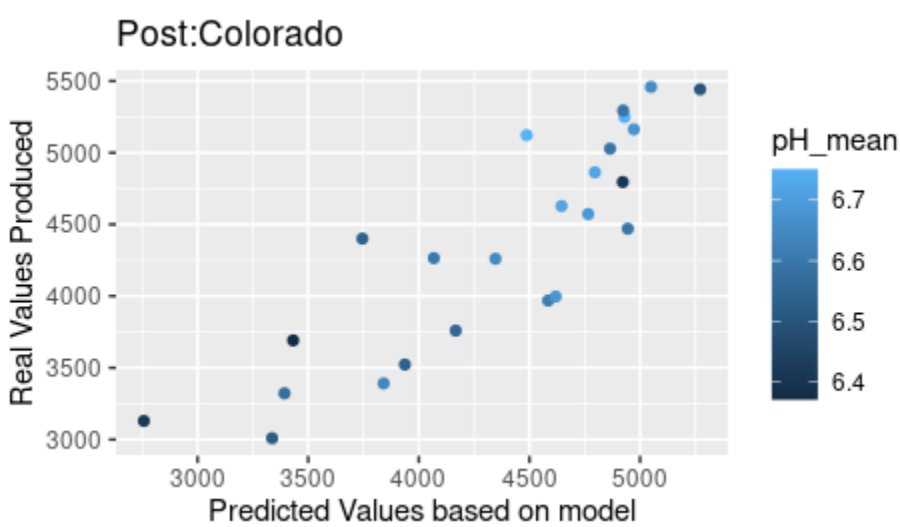

Figure 7: This graph is the predicted values of yield load adjusted from the data collected from colorado post-CAAA with the color of the data points being the pH of the data.

For Vermont the significant variables did not change but the coefficients of those variables did. For Vermont, the pH is 42100 in the post-CAAA data set when compared to pH in the

pre-dataset being 33200. In California the post-CAAA dataset led to 2 new significant variables in GPP and precipitation.

GLM

The lambda tuning via cross validation found that a minimal value is ideal for reducing error. Figure 8 shows that very small values for lambda give the lowest RSME. The coefficients created by a range of lambda values is presented in figure 9. While penalty values this low may lead to overfitting, for the purposes of our investigation (which does not involve prediction or extrapolation) minimizing RSME is the most important factor.

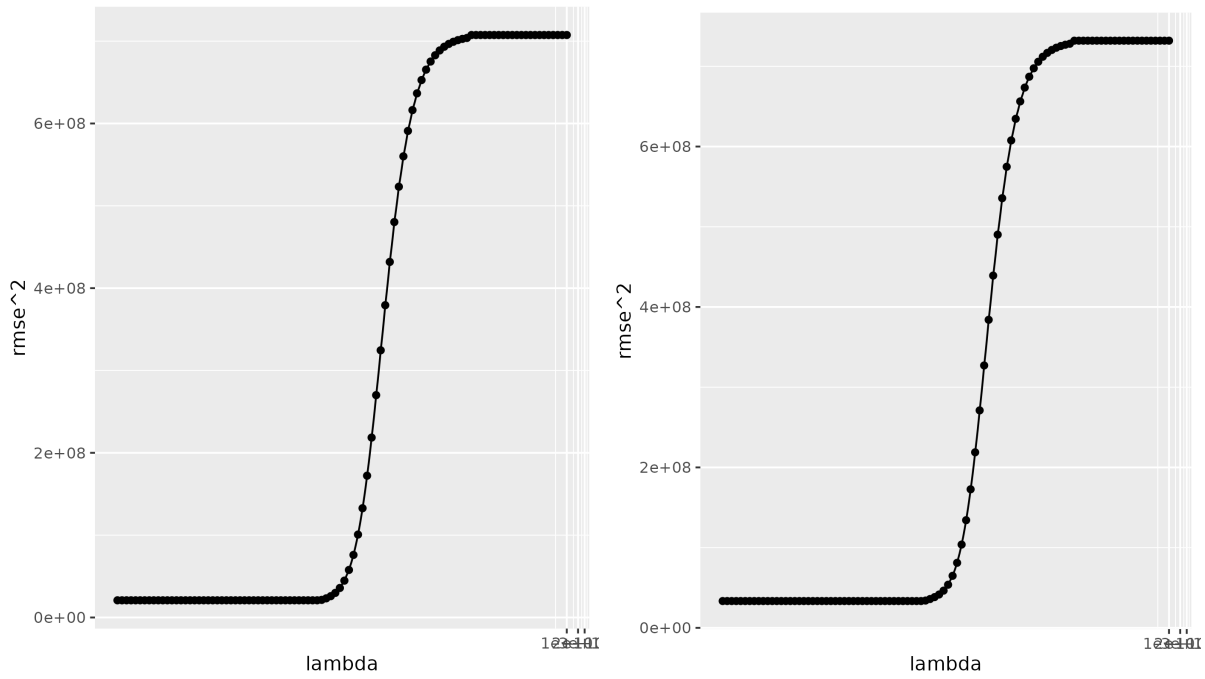

Figure 8: Results from cross validation for tuning lambda used in ridge regressions of generalized linear models. Pre-CAAA data is on the left and post-CAAA is on the right. Note that both perform best with a small value for lambda.

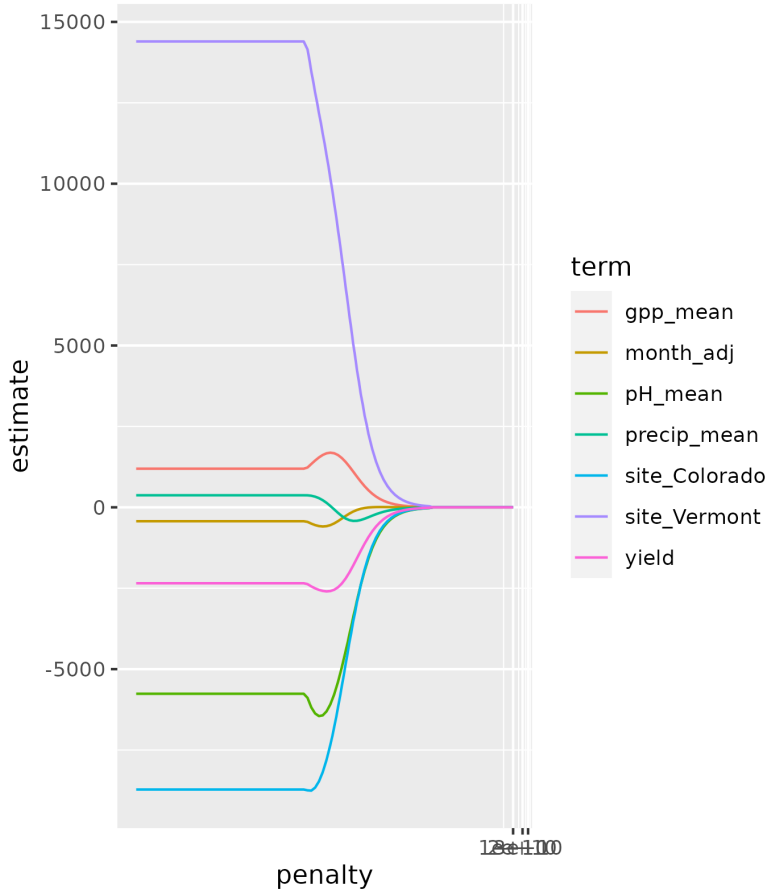

Figure 9: Example of coefficient estimates from ridge regression with changing lambda. The data used in this plot is from the pre-CAAA years. The post-CAAA plot looks about the same as this pre-CAAA plot.

Using the tuned lambda values, pre-CAAA and post-CAAA models were fitted. The resulting models both produced R-squared values of ~0.97. The values for these models are present in table 2. The only term that showed significant change was the precipitation term, which changed signs from a positive driver to a negative driver.

Table 2: Estimates of model coefficients on both datasets. Positive coefficients are shaded in green and negative coefficients are shaded in red. Note that while most coefficients stay about the same, the precipitation term changes from being a positive to a negative.

Term	Estimate Pre-CAAA	Estimate Post-CAAA
Intercept	25363.4107	26455.07137
Yield	-2346.1563	-2391.4788
pH	-5761.0476	-5446.3521
GPP	1193.7034	1744.0316
Precipitation	372.3861	-155.2876
Month of Water Year	-429.4684	-446.3582
Site Effect Colorado	-8720.0702	-6819.5748
Site Effect Vermont	14395.6741	15391.4097

Random Forest

One final method of variable selection/confirmation was in random forest's importance variable matrix. Using our final selected variables, random forest produced an importance matrix for both pre-CAAA and post-CAAA months (figures 10 and 11).

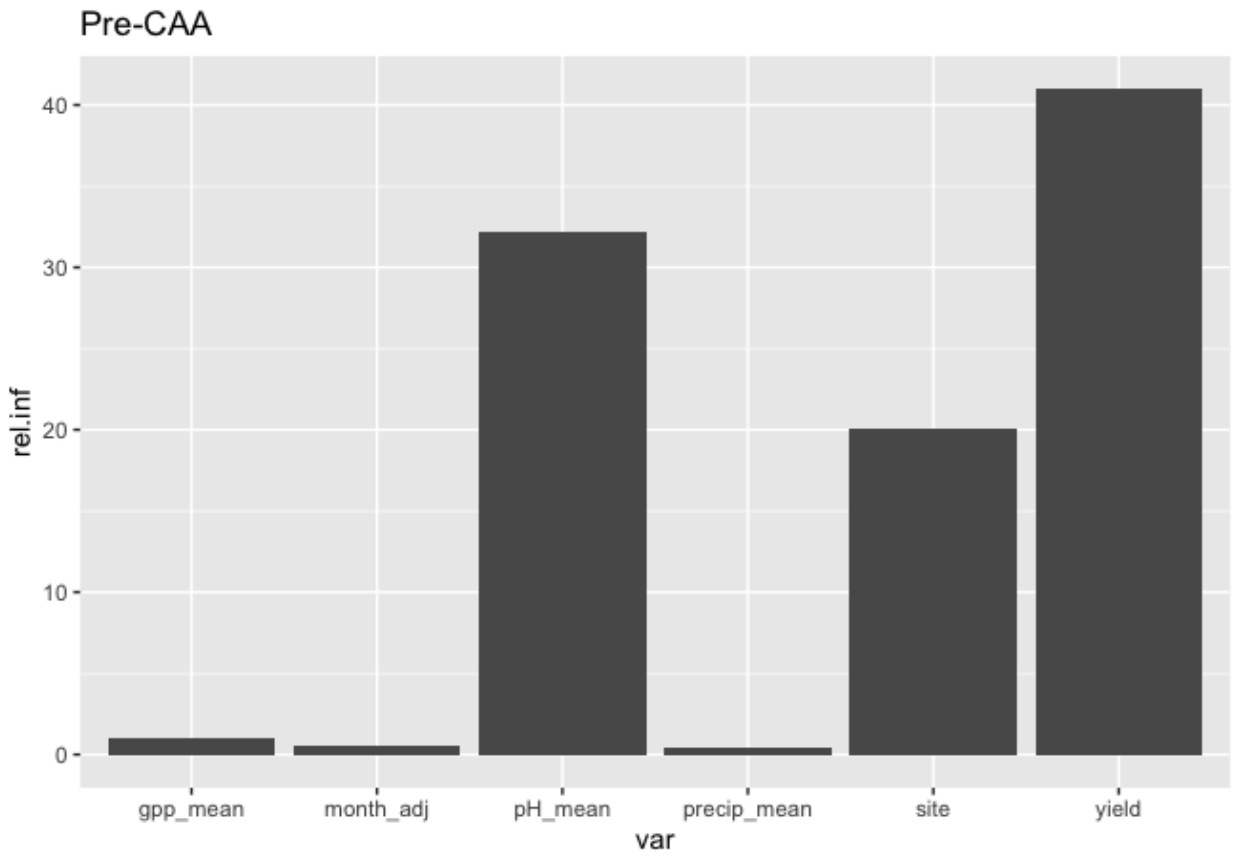

Figure 10: Random forest bar chart of variable importance for Pre-CAAA months at all three sites.

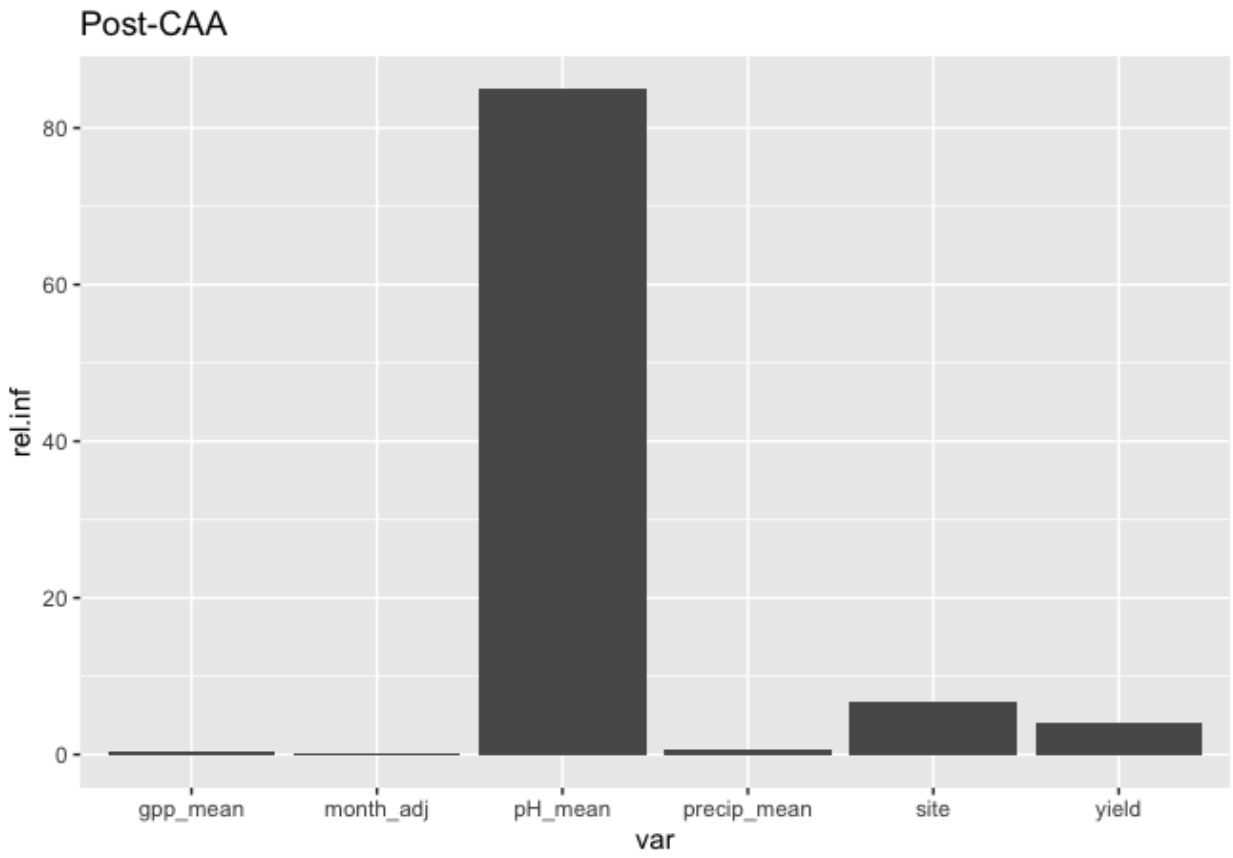

Figure 11: Random forest bar chart of variable importance for Post-CAAA months at all three sites.

Bagging results for pre-CAAA resulted in a training R squared of 98.36, and a test RMSE of 68027488. The results of this model can be seen in figure 12.

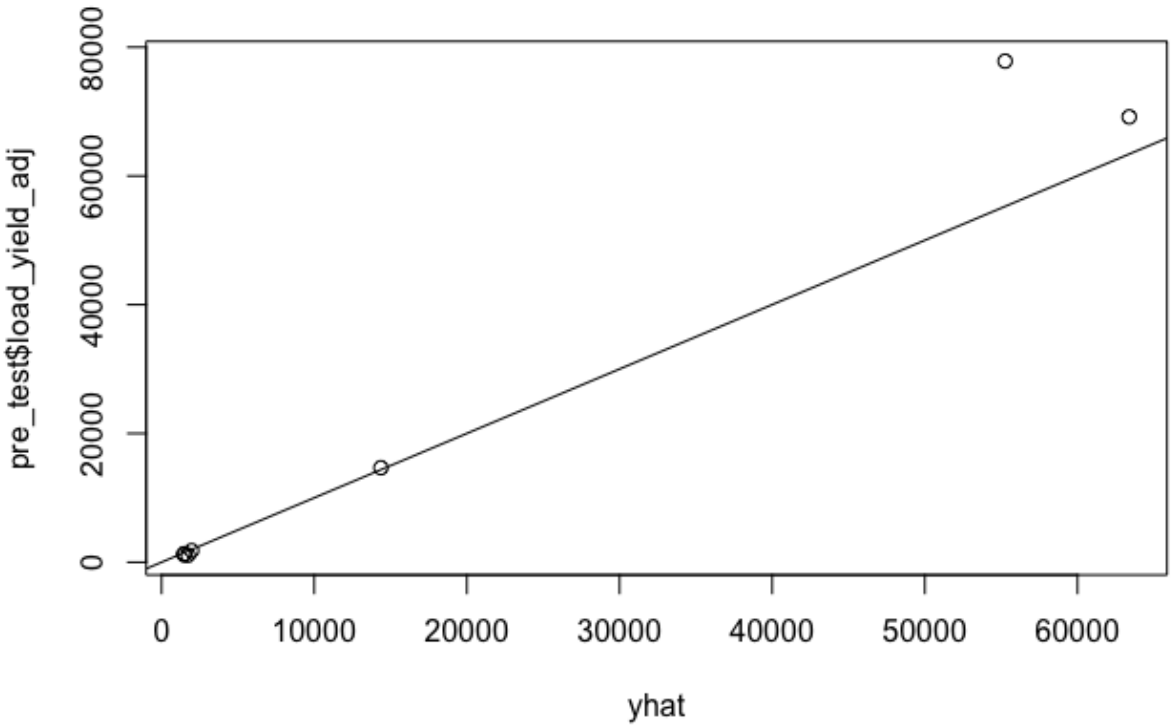

Figure 12: Predicted values and actual values of adjusted load from the random forest mode.

Bagging for post-CAAA resulted in a training R squared of 97.8 and a test RMSE of 15321280. The results of bagging can be seen in figure 13.

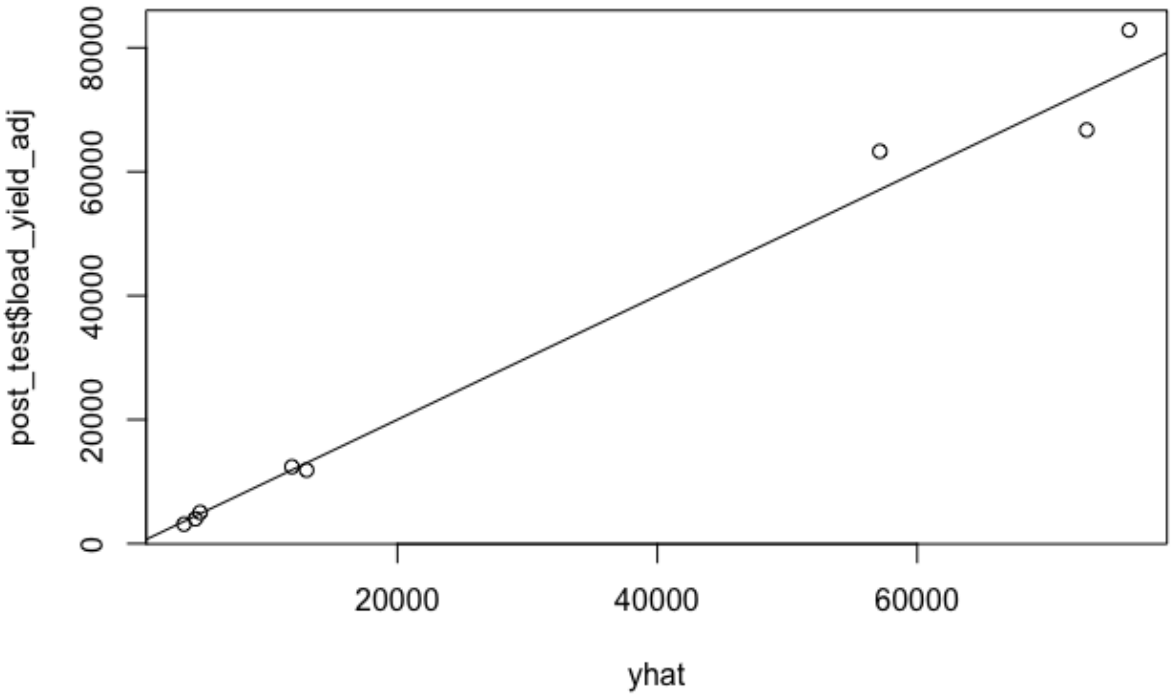

Figure 13: Graph of predicted values and actual values of adjusted load for random forest with bagging.

Gradient boosting significantly improved RMSE values, but less so for R squared values. Pre-CAAA testing R squared is 82.8% and RMSE is 2130.677 (figure 14). Post-CAAA testing R squared is 92.8% and RMSE is 8427.196 (figure 15).

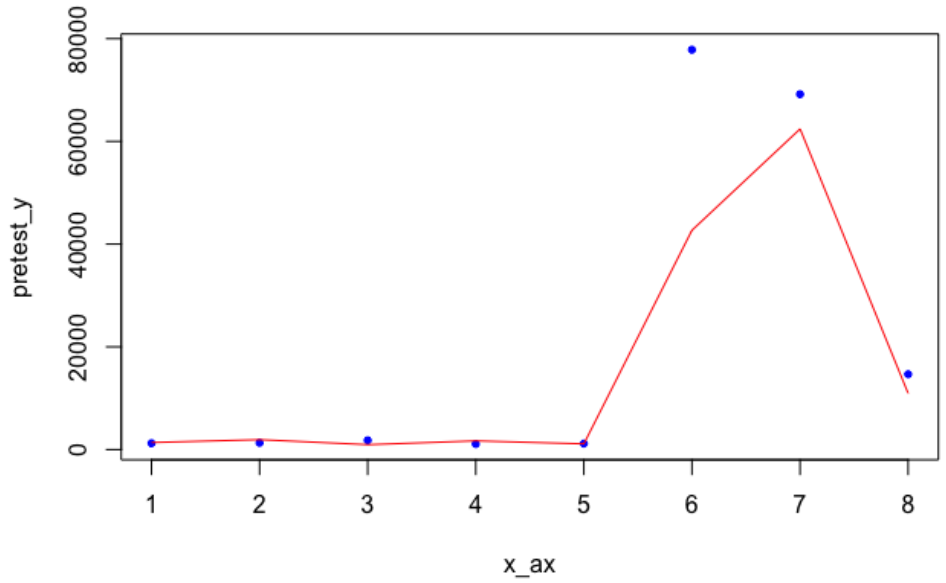

Figure 14: Graph of gradient boosted actual vs predicted data for pre-CAAA.

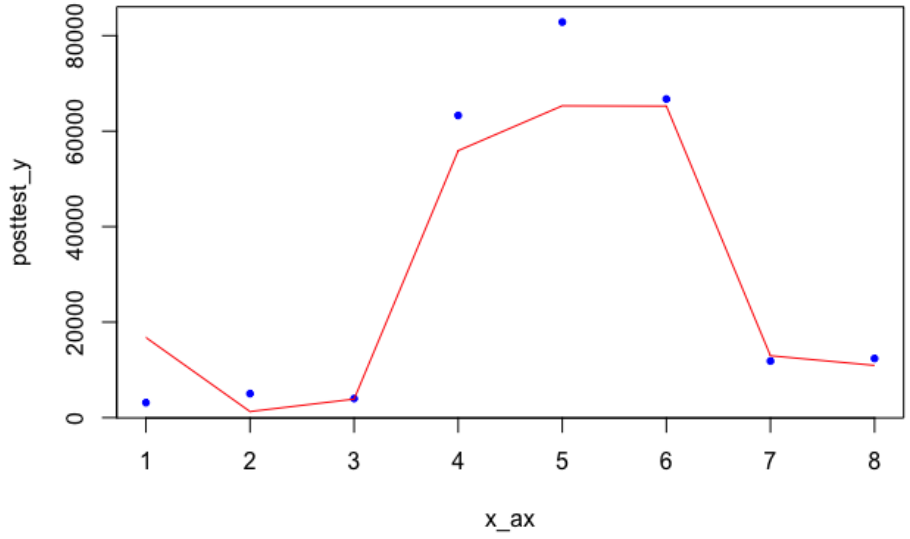

Figure 15: Graph of gradient boosted actual vs. predicted data for post-CAAA months.

Discussion

Together, all three model results give us information on how reduced acid deposition has changed the ways watersheds chemically weather. Our multiple linear regression models show a shift from acidity being the only good predictor, to other predictors (like GPP, precipitation, and yield) becoming useful. This fits with our understanding of how acid rain works. Initially, excess acid in these watersheds was the primary driver of chemical weathering. Since rainwater has become less acidic, other factors now exert more control on weathering processes across all of our sites.

This is confirmed more robustly with our ridge regressions. Comparing coefficient estimates from before and after the CAAA, we see most predictors have remained the same, but precipitation has shifted from being a positive predictor (more rain leading to more load per yield) to a negative predictor (more rain leading to less load per yield). This reflects a changing role for rain in the system. Prior to the CAAA, rain was a vector for acid delivery, enhancing weathering. Decades after the CAAA, we see that rain is a diluting factor. Reducing environmental acidity, and with it the rate of weathering across our systems.

These patterns are given some nuance by our random forest model results. In our pre-CAAA models, we see that pH, yield, and site are particularly important drivers. This is likely due to the disparity in historic deposition rates between the Eastern and Western US. The Eastern US has been historically more industrial and faced the most intense acid deposition prior to the CAAA (Stoddard et al, 1999). Decades after the CAAA, we see that the site effect drops significantly in importance, reflecting that all three sites are now under a more consistent regime of acid inputs.

All of all results point towards the importance of federal action on problems of dispersed pollutants. While our methods are limited in scope and robustness, they all show clear signs of how effective collective action can be in solving environmental issues.

Works Cited

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